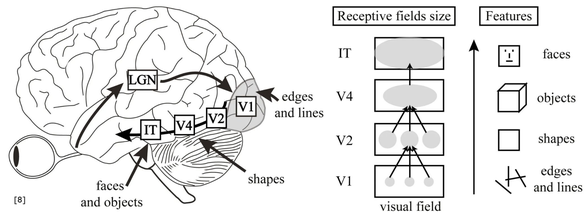
In Silico Mapping of Visual Categorical Selectivity Across the Whole Brain

Hossein Adeli[1], Ethan Hwang[1], Wenxuan Guo[1], Andrew Luo[2], Nikolaus Kriegeskorte[1]
[1]Zuckerman Mind Brain Behavior Institute, Columbia University, New York; [2]The University of Hong Kong, Hong Kong

Columbia University in the City of New York
Columbia | Zuckerman Institute
Visual Inference (Kriegeskorte) Lab



- Extensive neuroimaging experiments, especially functional magnetic resonance imaging (fMRI), have mapped a few prominent categories, faces, places, words, bodies, and food , to dedicated brain regions.
- But visual perception goes beyond simple categories and it remains an open question what higher-level visual concepts enable humans to make sense of the complex world.
- Here, we leverage recent advances in artificial intelligence and a large-scale fMRI dataset to explore the visual selectivity of brain regions beyond the well-studied visual cortex.
- We found many parcels with more complex selectivity transcending simple categorical concepts.
- Finally, we verify our labels by testing how well CLIP representations of our "superstimuli" can predict ground-truth activation on a held-out set.

Schaefer functional parcellation

We partitioned the 327,684 cortical vertices across the whole brain into 1,000 functionally homogeneous parcels using the Schaefer resting-state functional connectivity parcellation [9].

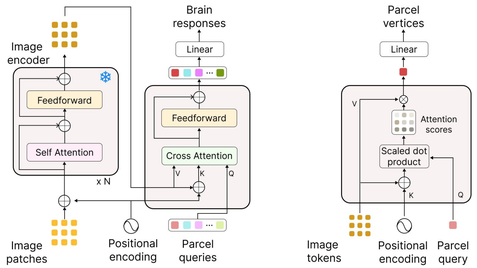
References

[1] E. J. Allen et al., "A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence," Nat Neurosci, vol. 25, no. 1, pp. 116–126, Jan. 2022, doi: 10.1038/s41593-021-00962-x.
[2] A. F. Luo, M. M. Henderson, L. Wehbe, and M. J. Tarr, "Brain Diffusion for Visual Exploration: Cortical Discovery using Large Scale Generative Models," Nov. 28, 2023, arXiv: arXiv:2306.03089. doi: 10.48550/arXiv.2306.03089.
[3] M. Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," Feb. 02, 2024, arXiv: arXiv:2304.07193. doi: 10.48550/arXiv.2304.07193.
[4] J. Deng, W. Dong, R. Socher, L.-J. Li, Kai Li, and Li Fei-Fei, "ImageNet: A large-scale hierarchical image database," in 2009 IEEE Conference on Computer Vision and Pattern Recognition, Miami, FL: IEEE, Jun. 2009, pp. 248–255. doi: 10.1109/CVPR.2009.5206848.
[5] H. Adeli, S. Minni, and N. Kriegeskorte, "Transformer brain encoders explain human high-level visual responses," May 22, 2025, arXiv: arXiv:2505.17329. Doi: 10.48550/arXiv.2505.17329
[6] J. S. Gao, A. G. Huth, M. D. Lescroart, and J. L. Gallant, "Pycortex: an interactive surface visualizer for fMRI," Front. Neuroinform., vol. 9, Sep. 2015, doi: 10.3389/fninf.2015.00023.
[7] J. Sergent, S. Ohta, and B. Macdonald, "Functional neuroanatomy of face and object processing: A positron emission tomography study," Brain, vol. 115, no. 1, pp. 15–36, Feb. 1992, doi: 10.1093/brain/115.1.15.
[8] M. H. Herzog and A. M. Clarke, "Why vision is not both hierarchical and feedforward," Front. Comput. Neurosci., vol. 8, Oct. 2014, doi: 10.3389/fncom.2014.00135.
[9] A. Schaefer, R. Kong, E. M. Gordon, T. O. Laumann, X.-N. Zuo, A. J. Holmes, S. B. Eickhoff, and B. T. T. Yeo, "Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI," Cereb. Cortex, vol. 28, no. 9, pp. 3095–3114, Sep. 2018, doi: 10.1093/cercor/bhx179.

paper

Project page

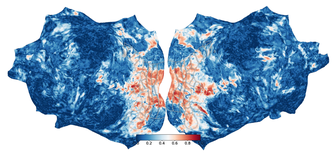
Transformer Brain encoder



Transformer Brain encoder (TBEn); Adeli et al (2025)[5]

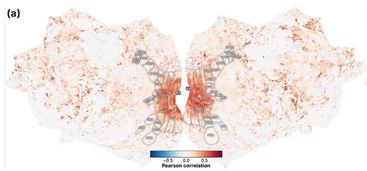
We built a transformer based brain encoder [5] to predict brain activity from visual features [3]. We trained the model on the NSD dataset [1], which includes up to 10,000 image-fMRI pairs for each of 8 subjects.

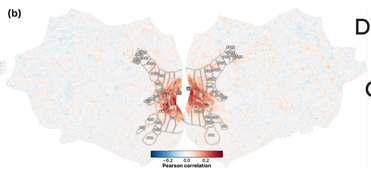
Prediction accuracy (pearson correlation)



As expected, our model performs well on predicting activity in the visual cortex. Several areas beyond the visual cortex are also well-predicted, which may offer insight into downstream visual processing.

TBEn vs Linear

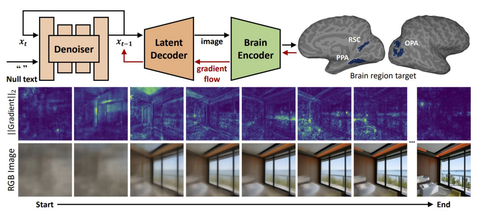




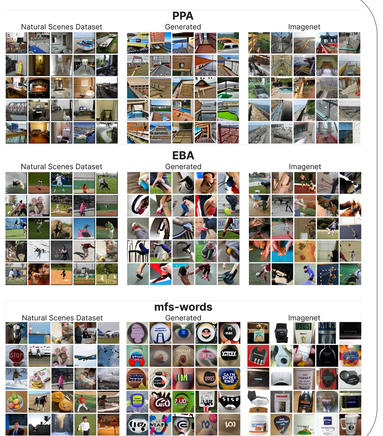
Dinov2 vs CLIP

Generating images with diffusion

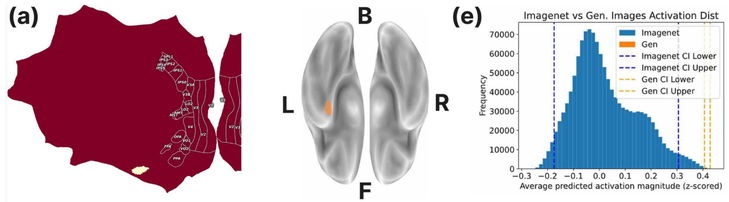
We used a diffusion based model, BrainDIVE [2], to generate images that would maximally activate a given parcel. The encoder model serves as a "digital twin" that is fully observable, upon which we perform extensive experimentation to better understand selectivity beyond the visual cortex.

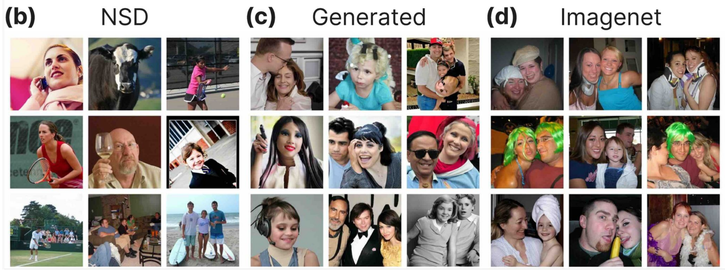


BrainDiVE; Luo et al (2023)



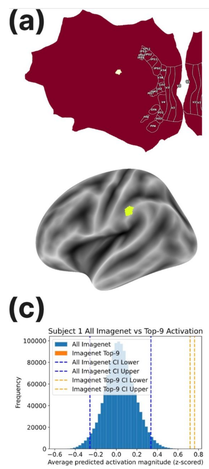
Sanity check: selectivity of known regions (aTL-faces)

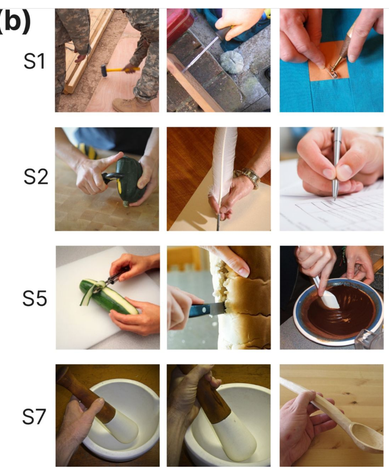




We start by demonstrating the selectivity of a parcel that overlaps significantly with a known region (a), aTL-faces, which is known to respond to faces [7]. We chose images from the NSD, BrainDIVE, and Imagenet [4] that maximally activate the parcel (b).

Demonstrating complex selectivity





"...utensils in action, often in practical, hands-on settings..."

"...hands actively engaging in a step of making food..."

"...objects or body parts being used as writing instruments or tools..."

"...action involving tools or sports equipment..."